

Integration and Generification—Agile Software Development in the Healthcare Market

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Abstract. The aim of this paper is to contribute to strategies applicable to vendors who want to move their locally designed and highly integrated systems to a larger market. A further aim is to explore how such systems developed for a local practice, and tightly integrated with the existing infrastructure, can be adapted to a larger market. We analyse the socio-technical mechanisms in play, the roles that the vendor and the users have in order to facilitate this, and the delicate interplay in relation to the other vendors. The analysis draws on the CSCW field, notions of *generification* of packaged software products, and boundary work. We argue that this process involves boundary work in relation to the installed base, as well as to other vendors and users. We also argue that the roles of the actors involved change during this process. The case described in this paper is the evolution of a system in which general practitioners can order laboratory services from the hospital electronically. The system integrates the general practitioners' information systems with the laboratory information system in the hospital. The development of the system started out in close cooperation with one customer, but as it evolved more customers bought the system. The system has been designed in an iterative and evolutionary way using agile development methods.

Key words: agile methods, boundary work, CSCW, generification, health care, integration, user roles

1. Introduction

Many software companies find that developing systems for a larger market is difficult. One reason often put forward is that systems are largely designed for a particular context, making it difficult to transport them to another (Hanseth and Lundberg 2001; Berg 1999). For vendors developing new systems, the key dilemma then becomes to determine how much local tailoring the vendor should offer, while trying to exploit the potential in the larger market at the same time by making the system as general as possible.

It is challenging in its own right to develop a system for a local work practice. Therefore, many software companies seek to exploit agile development methods such as Scrum (Schwaber 2004) and Extreme Programming (Beck 2000), which

include a high degree of user participation in order to map user needs. This is especially relevant for new development projects, which require mutual learning by users and vendors. Users learn about the possibilities of a system while vendors obtain knowledge about the users' work practices with the aim of ensuring robust use.

Often the development of new systems also presupposes a high degree of integration with the existing portfolio of information systems, the so-called installed base (Hanseth and Lundberg 2001). This is essential when the overall goal of the new system is to improve information and workflow between different organizations, departments, or professionals, where each of them already uses a domain-specific system. In this case, the success of the new system depends on how well it is integrated with the other systems, and how well the vendor of the new system manages to negotiate agreements with the other vendors. This is by no means easy, as the vendors may have different agendas and may pursue their interests from a more or less powerful position (Sahay et al. 2009).

If the new system works well, the next step might be to try transporting it to another context. The vendor will naturally strive to generalize features of the system in order to avoid maintaining several versions of it at different installations. However, the tight embeddedness with the local context and the tight integration with the existing information system portfolio, which initially ensured a well-functioning system, may now have the opposite effect, placing restrictions on generalization.

The paper contributes to strategies applicable to vendors aiming to move their locally designed and highly integrated system to a larger market. We analyse the socio-technical mechanisms in play, the roles that the vendor and the users have in order to facilitate this, and the delicate interplay in relation to the other vendors.

The case studied in this paper is the development of a system for electronic ordering of laboratory services made by the vendor Well Diagnostics. The system, which is named Well Interactor, enables general practitioners (GPs) to choose and order laboratory services directly from their computer. From the first working version, Well Interactor needed to be integrated with two existing information systems. One was the GPs' electronic patient record, and the other was the laboratory infrastructure in the university hospital involved. The development of the system started out in close cooperation with one customer, but as it evolved, more customers bought the system. Well Interactor has been designed in an iterative and evolutionary way using agile development methods.

This raises the question of how systems developed for a local practice, and tightly integrated with the existing infrastructure, become applicable to a larger market. In our analysis of the case, we first explore how the vendor Well Diagnostics negotiated the boundaries of its (initially non-existent) product with regard to the systems from the other vendors, and how they later increasingly created space for themselves as their product emerged successfully. Secondly, we

examine how the vendor coordinated the large user groups from different contexts throughout the process and how the agile methodology shaped the process. Thirdly, we analyse how the roles of the actors actually changed as the vendor obtained a larger market share.

We draw on the CSCW field, which has a long tradition of paying careful attention to the users' perspective in local practice by highlighting the gap between design and use (for instance, see Berg 1999; Schmidt et al. 2007; Ellingsen and Monteiro 2003). While this literature has convincingly demonstrated the value of focusing on micro-practices, less attention has been paid to how systems in local practices scale up (Schmidt and Bannon 1992), essentially, making a local system work in other contexts. We supplement our framework with the notion of *generification* of packaged software products (Pollock and Williams 2008) which is part of the vendor's strategy of making a system work in several settings. We also draw on the notion of boundary work (Gieryn 1999; Barrett et al. 2007; Hernes 2004) in order to elaborate on the negotiations and compromises related to the other vendors and the users.

The paper is organized as follows: The next section describes the theoretical framework and concepts used in this paper. The methods section describes and explains the methods for the empirical research. The sections named 'Motivation and Status' and 'Case' contain the empirical findings presented as the history of the product being studied. These sections are followed by an extensive discussion of the findings in the case.

2. Theory and concepts

The CSCW literature has focused extensively on how information systems or artefacts can support distributed collaborative work among groups of users (Berg 1998; Carstensen and Sørensen 1996; Egger and Wagner 1993). Key questions in this research have then been to suggest suitable design methods and how to include the users in the development process (Heath and Luff 1992; Berg 1998). Often, the technology in question has been recognized as being singular or standalone, like paper artefacts, shared information displays and monitors (Heath and Luff 1992), isolated information systems, concrete products such as Lotus Notes or 'primarily off-the-shelf products' (Hartwood et al. 2002; Berg 1998; Grudin 1994).

This reflects a general trend in the CSCW tradition to focus on the activities of small groups in local contexts (Aanestad 2002; Bannon 1998, Grudin 1994; Carstensen and Sørensen 1996). This has also been observed and criticized within the CSCW community itself. As Bannon (1998, p.41) underscores, the focus on small teams "*pay[s] little attention to settings in everyday organisational life where issues such as power and politics play a large role*". A related problem with focussing only on singular artefacts is that many information systems development projects currently presuppose integration with existing systems to

ensure efficient information and process flow across institutional, departmental and professional boundaries. For software companies developing new systems, this raises two important questions. The first is how they can integrate their software with existing IT portfolios, and how to negotiate with the vendors of these systems. The second question is how their system, once implemented in a local setting, can be dismantled from existing integrations to be moved to a completely new context.

Firstly, creating a new system which is highly integrated with an installed base requires the designers to have relatively detailed knowledge about the existing systems, interfaces and work practice. The designers must take into account interdependencies between the systems, information flow, different technologies, responsibilities, interests and layouts, etc. Ellingsen and Monteiro (2007) illustrate this in a case from a microbiology laboratory which wanted to decide exactly how the results sent from the laboratory system should be presented to GPs in their electronic patient records (EPRs). Another example provided by Johannessen and Ellingsen (2008) describes how the same laboratory wanted the costs of the potential laboratory analyses to be presented in the EPR when GPs made online orders to the laboratory, to discourage the GPs from ordering costly analyses that were assumed 'unnecessary'. Thus, designing a system to form part of a large information infrastructure requires thorough local knowledge and tailoring that extends beyond the scope of the new system itself. This also makes it evident that Bannon's argument (quoted above) about power and politics is more relevant than ever.

Secondly, an essential question is how an infrastructure-like system, carefully designed for one local context, can be moved to another. This is often of great concern to vendors and designers who have invested substantial funding and resources in developing a local system, where the effort and resources invested in the first version of a product often exceed the payment from the customer. Particularly for small vendors of information systems, the ability to take a product made for one customer into a broader market may be a question of survival. However, as studies in CSCW and STS have informed us, this is not an easy task. Marc Berg (1999) claims that many systems in healthcare are failures because they were not transportable out of the local context. Along similar lines, Hanseth and Lundberg (2001) claimed that in organizations with particularly complex work practices such as hospitals, only the users themselves would know the practices well enough to be able to design a system for that organization. When we also take into account the added complexities involved in existing integrations, the effort of moving a system to a new context may seem insurmountable. Still, in spite of the reported difficulties, some systems have been successfully transferred across organizational boundaries and implemented in many organizations. Some examples are global company-wide information systems (Rolland and Monteiro 2002) the global Enterprise Resource Planning (ERP) system SAP (Pollock and Williams 2008) and large-scale EPRs

implemented in several hospitals. Based on such evidence, Pollock et al. (2007) claim quite correctly that the STS literature is biased in the way it has focussed on the gap between the functionality of a system and a local work practice in the implementation of a new system:

“In our view, the STS literature tends to over-emphasize the collision between specific organizational practices and generic system presumptions at the point of implementation within specific user organizations (...) This, we would argue, reflects the various narrative biases within current STS and sociology: that contexts of use are always individually different, unique and typified by highly idiosyncratic practices; whereas technologies are ‘singular’ and ‘monolithic’; and localization is the means by which the standard and the unique are somehow brought together” (Pollock et al. 2007, p.257).

Pollock and Williams (2008) have analysed how ERP systems are implemented in many organizations worldwide, and suggest that we should focus on how these systems are made ‘general’. They introduce the notion of *generification*, which is “*the supplier strategy of taking a technology that has worked in one place and attempting to make it work elsewhere, and, in principle, everywhere*” (Pollock and Williams 2008, p. 129). Central in a generification process is the trade-off between particularization and generification. Typical questions include: what are particular requirements from a few customers and what are general ones, how particular are those requirements from the few customers, should diversity be built into the system or should functions meeting these particular needs be customized at each site?

While we have great sympathy for the perspective introduced by Pollock and Williams (2008), our study differs in two principal ways. Firstly, their account of generification focuses primarily on the implementation and adaptation of large scale ERP packages. Often vendors of such packages attempt “*to avoid getting too close to a specific local context*” (Wang 2007, p. 108). Combined with the fact that such packages often are built on existing packages or local systems (see Pollock et al. 2003, p. 323; Wang 2007, p. 103), this limits the potential for adaptation. In comparison, the product developed in our case is completely developed from scratch. The development process has involved an agile methodology, presupposing heavy user involvement and delivery of working functionality early in the development phase. We argue that trying to ‘move out’ such a highly locally-tailored system is much harder.

Secondly, by their nature ERP systems have a company-wide scope, and therefore integration with other systems is not an absolute condition (although it is desirable). In our study, integration is an essential condition for a working product. Integration with the larger information systems portfolio must be in place right from the first release of the software. Therefore we are particularly concerned about how vendors of new systems negotiate with vendors of the existing systems who are major players already (for instance vendors of EPR

systems) in the healthcare segment. This negotiation may be performed from positions of varying influence. Vendors of new systems may find themselves in an inferior position where the vendors of the existing systems typically *prescribe* how the access to their system should be handled (Web services, XML, etc); the ‘rules of engagement’ so to speak, while the vendor of the new systems will have to *perform* the actual integration. Such integration expresses an unbalanced relationship between the participating vendors, or ‘asymmetrical integration’ (Sahay et al. 2009).

Developing a system from scratch with the overall goal of gaining market shares entails a great deal of negotiation between the ‘particular’ and the ‘general’. This includes establishing and maintaining the boundary between the particular and the general functionality, between the particular and general user (Pollock and Williams 2008) as well as between the new system and the installed base (hence also the other vendors). These boundaries are not fixed, but will be renegotiated and redefined (Hernes 2004; Barrett et al. 2007) as the development process emerges and the system is shaped and moved between contexts. The task of establishing and maintaining these boundaries resembles what many researchers in the STS field refer to as *boundary work* (Gieryn 1999; Burri 2008; Barrett et al. 2007). The concept is widely used in organizational settings to describe the strategic behaviour and the circumstances related to engaging in work to sustain boundaries between different communities of practice. “*Generally, when privileged groups engage in boundary work it would mean (...) constructing and maintaining distinctions between themselves and others*” (Barrett et al. 2007, p.8). For a vendor aiming to develop a new system, this might include negotiating the boundaries with the larger information systems portfolio or clarifying the scope of the system in relation to the users. However, what is particularly interesting in developing new systems is how these boundaries are expanded (Barrett et al. 2007; Gieryn 1999) due to added functionality, integrations, increasing numbers of users and new requirements. In this sense, we may observe an increasingly influential role of the system, which is in accordance with Lee’s (2007) argument that “*artefacts can be used to push boundaries*” (2007, p.308).

3. Method

The research was mainly carried out in the company Well Diagnostics and the in University Hospital of North Norway (UNN). Well Diagnostics is a small company with 15 employees specialising in systems for communicating and interaction across organisational borders in Norwegian healthcare. The product portfolio includes products for secure communication between patients and healthcare and between different organisations in healthcare. It also includes software for capturing and storing medical multimedia information and for booking patient appointments. Well Diagnostics works in close cooperation with

vendors of different hospital and primary care information systems with which the products of Well Diagnostics need to be integrated. UNN is the largest hospital in the northern region of Norway, with approximately 5,000 employees and 600 beds. The hospital has seven laboratories: clinical biochemistry, microbiology, pathology, clinical pharmacology, immunology, the blood bank and medical genetics.

The study adheres to an interpretive research approach (Klein and Myers 1999; Walsham 1995). The analysis of the data is based on a hermeneutic approach where a complex whole is understood “from preconceptions about the independent meanings of its parts and their interrelationships” (Klein and Myers 1999, p.71). This implies that the different sources of field data are all taken into consideration in the interpretation process. The method included relatively detailed case write-ups for the sites involved (see for instance Eisenhardt 1989) followed by an examination of the data for potential analytical themes. Methodologically, the authors are influenced by the concept of biography coined by Pollock et al. (2003, 2008). The biography approach studies artefacts as heterogeneous assemblages, and follows the accumulated history of an artefact in its historical context and across its life cycle. The life cycle of the product involves many stages from birth through implementation, and the biography focuses on the transformations and translations that a product goes through as it is changed and redefined during its life cycle.

“The biographical approach focuses upon social (or rather socio-technical) processes involved in innovation and how these are shaped by their context and history. [...] Multiple methods may be required, knitting together different kinds of evidence including historical studies, ethnographic research, qualitative studies of local and broader development and the use of larger-scale research instruments and quantitative data” (Pollock and Williams 2008).

Although Pollock et al. have intended this biography concept for a class of systems, not a particular product, and over a longer time span than in our case, we find the concept useful for framing our research.

Data gathering conducted by the authors consisted of participant observations (work settings and project meetings), interviews and informal discussions. In the period from December 2007 to March 2009 the first author participated in approximately 60 meetings related to electronic requisitions, lasting from 15 min to 6 h. These were internal meetings at the laboratories, information sessions with GPs and secretaries, guided tours in the laboratories, risk analysis meetings, training sessions, meetings and workshops between the user groups and internal meetings in the development team. The meetings involved a broad range of professions: IT consultants, administrative personnel, physicians, GPs, biomedical laboratory scientists, marketing personnel and developers. Notes were taken during these observations and subsequently refined and organized. Questions and

analytical points were added ex post. The first author had an office in Well Diagnostics allowing her to participate in informal discussions, for example during lunch breaks, which facilitated awareness of emerging situations and issues. The findings have been discussed with the second author who has a thorough knowledge of the field from studies of innovation in health care in general and information systems in hospital laboratories in particular.

In addition to the meetings, we were particularly interested in the history of the project, from the time before the authors were involved, and also the reflections and ideas of some of the key people involved in the process. The authors therefore conducted nine in-depth semi-structured and unstructured interviews, lasting from 30 to 105 min, comprising one with a physician, one with a biomedical laboratory scientist, one with the project leader for electronic requisitions at UNN, one with the person responsible for electronic requisitions at the University Hospital of Akershus, one with a manager at the collaborating vendor Profdoc, three with designers in Well Diagnostics (twice with the same person) and one with the person in Well Diagnostics responsible for the marketing of Well Interactor. The first author also conducted one semi-structured group interview with general practitioners and their laboratory personnel.

The first author is employed by Well Diagnostics as a PhD student. The research project was financed partly by the research programme 'Tromsø Telemedicine Laboratory' and Well Diagnostics in a joint effort to integrate research and industry. The first author has presented her findings both in informal discussions in Well Diagnostics and by letting the subjects read written accounts during the course of the research. This has forced reflection and further discussions by both informants and researcher.

4. Motivation and status

In 2001 the University Hospital of North Norway (UNN) studied the use of resources and error rates related to their own laboratory activities. This study revealed that the paper-based requisitions from primary care often contained errors, lacked clinical information or had a mismatch between the paper-based requisition and the sample tube. In addition, manual and repetitive work in receiving the samples was considered a waste of resources. To assure quality in the reception of requisitions and samples, UNN saw the potential for an electronic system for requisitions from the GPs. At the same time Well Diagnostics had an idea for a system for electronic requisitions that was fully integrated both with systems in general practice and in the hospitals.

"I understood that we could make a system outside of the GP's electronic patient record. [...] But we were lucky with the timing. At the same time UNN had done their study of the organisation of the laboratories, and found that they could be more effective by doing things in a different way. Then we hit the

target; we had a customer, and then the ball started to roll.” (Chief Designer, Well Diagnostics).

As a result, UNN and Well Diagnostics established a project, GiLab, aimed at designing a system that enabled the interaction between general practice and the laboratories. By committing themselves to cooperation in the development of the product, they received approximately 30% of the total budget from Innovation Norway, a national institution supporting industrial development. One of the main strategies of Innovation Norway is to support the developments of products made for one customer, aiming to eventually selling it to a larger market. Innovation Norway considered that the planned system had the potential for being taken further to eventually reach a larger market. This was in line with the ambitions of Well Diagnostics as well. Software development is labour intensive and costs for packaged software are high (Campbell-Kelly 1995), hence selling the product to more customers would be a possibility to increase the payback for the product. For UNN it was beneficial in the sense that the system would be further developed and costs could be shared between several customers. The project was ready to start in the beginning of 2006.

The GiLab project phase lasted for approximately 2 years. Four GP offices in the area around UNN were piloting the system from the start. The numbers of users are increasing and presently 13 offices around UNN and a total of 26 GP offices around other Norwegian hospitals are using the system for electronic laboratory requisitions. An extension of the system to interactive referrals with booking is in the process of being implemented in all 150 GPs in Troms County, and interactive referrals to Akershus University Hospital will be implemented soon at four GP offices, see Table 1.

5. Case

This section presents the different stages of the evolution of Well Interactor. It is important to mention that the different stages presented are not completely chronological, but that they partly existed in parallel. For instance, the system was already sold and implemented at the University Hospital of Akershus before

Table 1. Timeline of the development of Well Interactor.

Early 2006	Innovation Norway decided to support the GiLab project
June 2006	Well Diagnostics wins the contract with Akershus University Hospital
September 2006	Start of the clinical biochemistry pilot at UNN
Spring 2007	Start of the generification to the Microbiology laboratory
September 2007	First pilot for Well Interactor at Akershus University Hospital
Fall 2007	Pre-project Innlandet and Asker and Bærum hospitals
Fall 2008	Start of the microbiology pilot at UNN
Spring 2009	Start of project for interactive referrals

the product was extended to different work practices, meaning that the stage of 'moving out of the local context' started before the stage 'generification across different laboratories at UNN' was finished.

5.1. Establishing the first local solution for UNN (2006)

5.1.1. *Close collaboration with users through agile methods*

Well Diagnostics has chosen to employ agile development methods throughout the company. Agile methodology is a conceptual framework for software development that has evolved in part as a reaction against 'heavyweight' methods like the waterfall model (Beck 2000). When traditional waterfall methods are seen as bureaucratic and slow, characterized by a pre-planned process of requirements capturing, analysing, designing, coding, and testing, agile methods are seen as the opposite. The main idea is that short iterations make the methods receptive to changes in the environment. The basic values for agile methods are expressed in The Agile Manifesto: *"Individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation"* (Beck et al. 2001). When Well Diagnostics started to build a system for electronic requisitions from scratch, they faced many unknown factors. Initially they had some ideas for the technological concept, but the developers did not know the work practice of the users well enough to be able to develop the system on their own. They had also experienced earlier that users were not usually able to formulate their needs and requirements well enough. Well Diagnostics therefore saw an agile approach as a way to make up for these shortcomings:

"We have years of experience of waterfall methods which have repeatedly failed. It does not work with a big plan in advance. And the customers are not able to formulate their wishes and needs clearly enough. I could make a long list of issues related to reaction to change and complexity that makes it necessary to do things in small steps." (Chief Designer, Well Diagnostics).

In their agile development approach, Well Diagnostics have borrowed from the agile methods Extreme Programming and Scrum, and adapted these methods to their own environment. Scrum focuses on project management in situations where it is difficult to plan ahead, while Extreme Programming focuses on best practice for development, for instance by being responsive to changes in the environment and developing only what is needed at that time.

An important value pervading agile methodology is the close relations between developers and users or customers. An agile approach implies that the developer gives high priority to satisfy the customer through early and continuous delivery of valuable software where changes of requirements are welcomed. Several professions both in primary care and in the hospitals constituted the user group of

Well Interactor. In general practice the users included GPs and their secretaries and local laboratory personnel, and in the hospital laboratories the users included physicians, biomedical laboratory scientists and computer support personnel. The design process of the early versions of Well Interactor was a collaborative effort between Well Diagnostics and these user groups.

“It is just as impossible for them to communicate their needs in technical terms as it is for us to transform the technology into their needs. So that is why this [the agile approach] is an efficient way of cooperating. If you deliver software frequently you get corrections all the way to get you onto the right track.” (Designer, Well Diagnostics)

Four months into the project, and in line with agile philosophy, Well Diagnostics presented a simple, but working solution to the customer:

“We kept the initial discussions as short as we dared. We took a chance delivering functionality early, and let the pilot GP practices [the users] shape the solution together with the laboratories. I think that was smart. [...] Make it as simple as possible to illustrate the intentions. When the users start using it they will see how this suits their daily work, and they will correct us and give feedback on how it should be.” (Designer, Well Diagnostics)

The developers at Well Diagnostics worked in three-week iterations. The first step in an iteration was to collect user stories and estimate the work involved in fulfilling them. According to agile methodologies, user stories are to be written and prioritized by the customers and they serve as a communication channel between developers and customers. In the design of Well Interactor the developers were faced with many user groups, each with different usage of the system and limited time to spend on design. Well Diagnostics therefore had to be more pragmatic in the writing of user stories. Although the user stories were based on requests or feedback from the users, they were formulated by the development team, and the actual users were not present during this process in the development team.

5.1.2. *The Well Interactor concept—the first solution*

The very first solution satisfied the minimum requirements to enable submission of an electronic requisition and was more or less a digitized version of the existing paper-based requisition. Based on the real-life use and testing of this solution the iterations continued and the product was continuously changed and adapted to the needs of the users. An example of such changes was the broadening of the scope of the product by including the receptionists in general practice as a new user group. Initially, only the doctors and the laboratory personnel were seen as users, but after the product had been used for a while the developers realised that much work is done by the receptionist, without the

patient actually seeing the physician. This resulted in new functionalities that accommodated for the receptionists' work.

The result of this process was the product named Well Interactor. The concept includes components to be used either on the GP side, on the laboratory side, or in the communication between them. The figure below shows the information flow for the Well Interactor concept. The concept includes: the existing information systems used by the actors involved in the GP practices and the hospital laboratories, the existing communication channels and standard Electronic Data Interchange (EDI) for exchange of information, and the components made by Well Diagnostics; Well Interactor Smart Client at the GPs, the SPP web hotel and a site server in the hospital, see Figure 1.

The smart client is fully integrated with the GPs' EPR and enables GPs to choose and order laboratory services directly from the laboratory's list of offered services, called the service provider profile (SPP). This list is unique for each single hospital and contains all information on the analyses and specimen requirements. The SPP also specifies what kind of information the laboratories will need from the ordering GP to execute the analyses. For GPs, this means that only a few extra clicks are needed to produce and send a laboratory requisition while they are working with the EPR. The requisition is then sent directly into the laboratory information system. Well Interactor creates labels with a barcode to be stuck onto the sample tube. When the samples are received at the laboratory, the barcode is scanned and information from the barcode is matched with the requisition.

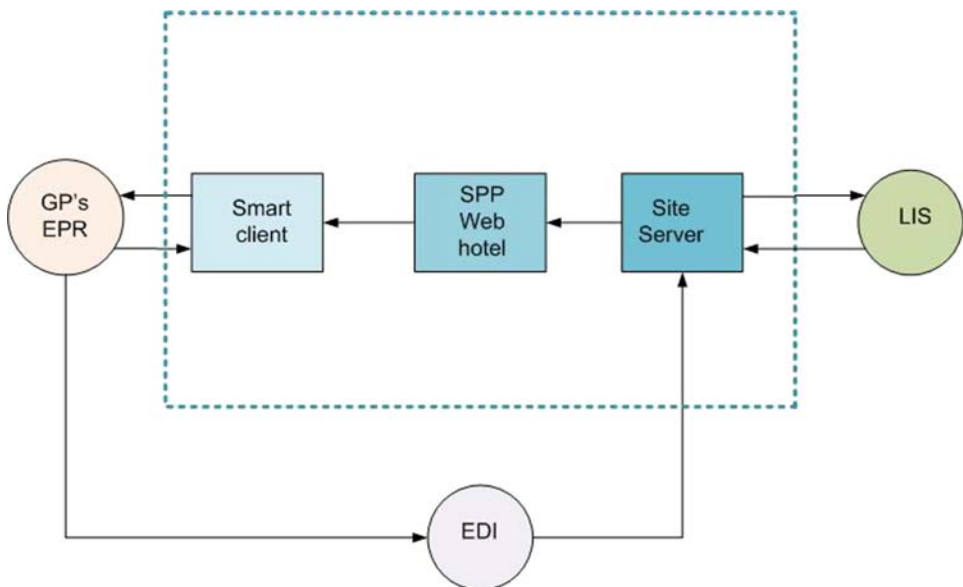


Figure 1. The information flow of the Well Interactor concept.

5.1.3. *Integration with systems from other vendors*

By connecting the EPR system at the GP practice and the laboratory system in the hospitals Well Interactor might be seen as a glue between these systems, making Well Interactor one out of several systems in a large portfolio which have to interact with each other. All from the beginning the idea was that Well Interactor should be very tightly integrated, or bundled, with the EPR system to make the work processes for the GPs as smooth as possible, not forcing the GP to switch between systems. This made it crucial to establish a close relationship with the vendor of the EPR system. The vendor of this system is Profdoc, a leading vendor of EPR systems for the Norwegian primary care sector having approximately 80% of the national market share. Well Diagnostics and Profdoc have a history of cooperation; Well Diagnostics has previously made components for Profdoc's systems, and Profdoc is selling one of Well Diagnostics' products as an extension to their own EPR system. This time Well Diagnostics was creating a system that would be a separate product, not necessarily seen as profitable by Profdoc. To ensure commitment as well as sharing risk Profdoc was invited as partner in the original GiLab-project, and granted a share of the Well Interactor product, hence giving them income from the sales of the product. The integration is partly regulated through national standards, but these have been deficient in specifying the content of electronic requisitions and this have posed challenges related to integration towards the GPs' EPR. Because of the lacking standards Well Diagnostics and Profdoc had to agree on the how the integration was to be executed, what information to exchange between the systems etc. The negotiations resulted in the definition of an API (Application Programming Interface), a kind of contract regulating the exchange of information between the smart client and the GPs' EPR. For instance the API describes how they will make up for the lacking content of national standards by using 'other' and 'miscellaneous' fields.

On the hospital side the integration work has been related to transporting messages into and exporting information from the hospitals information system. The first version of Well Interactor was integrated with the laboratory information system DIPS Lab. DIPS ASA is the largest Norwegian vendor of hospital information systems. All information required for the SPP was already in DIPS Lab, but the system was not able to export this information. This implied total manual creating and editing of the SPP done by the developers in Well Diagnostics. To be able to import the requisitions into DIPS Lab another API had to be defined between the Site Server and the hospital information systems, among others to clarify the use of 'other'-fields as described in the other API.

The establishment phase of Well Interactor resulted in a local solution. This solution was tailored to the particular needs of the laboratories involved at UNN

and the four pilot GP practices as well as their corresponding laboratory information and EPR systems.

5.2. Generification across different laboratories at UNN (2007)

In accordance with agile methodology, the development had started at one end, integrating with one laboratory system and one EPR system. Now it was time for the product to expand, and the generification process at this stage was related to the different needs that emerged as the microbiology laboratory at UNN was included. This implied adaption to new work practices as well as integration with another laboratory information system.

5.2.1. *Different work practices and the development of an editing tool*

Well Diagnostics experienced that the microbiology laboratory work processes were different than the practices of the clinical biochemistry laboratory. The differences appeared in analysing techniques as well as in the logistics of handling the specimens and the information related to them. While much of the work in the clinical biochemistry laboratory was routine work with large quantities of analyses, the volumes at the microbiology laboratory were much lower and many analyses were carried out in a manual manner. When ordering analyses from the clinical biochemistry laboratory, a GP would usually know exactly which analyses to order to get the answers she needed. The laboratory would then only need minimal, if any, clinical information from the ordering physician. The process of ordering from the microbiology laboratory on the other hand requires extensive clinical information. An example could be a patient with stomach problems after a vacation in another country. The GP would then write this information in the requisition and ask if there could be any bacteria or parasites in the specimen, and if so, which. In addition to the test results, the response would then include an interpretation and explanation of the results, and possibly also recommendations for treatment. These differences in work practices resulted in different needs for the information content of the electronic requisitions. The first SPP was a long searchable list of the analyses. As the microbiology laboratory's analyses were included, the list became too long and there was a need to organize the list differently. So far all editing of the SPP had been done by the developers, but with more laboratories and more frequent editing, the need to make editing of the SPP easier emerged. An editing tool, the SPP Publisher, was therefore developed making it possible for the hospital personnel to be able to create or edit the SPP in an easy manner themselves. The information system at the microbiology laboratory could easily export all needed information to the SPP Publisher (while the clinical biochemistry laboratory still had to do this manually). In the SPP Publisher the presentation for the GPs could

then be made, and additional information could be added. Every time changes was made in the laboratory's SPP a new version was uploaded to the SPP web hotel, and from there the GP's Smart Client was automatically updated.

Well Diagnostics as well as the SPP editors at the laboratories also experienced different work practices within the group of GPs. The SPP appeared as a choose-and-pick list for the GP. The way the SPP is structured influence the efficiency of the GPs' laboratory ordering process, so the laboratories put much effort into making good SPPs. A discussion among GPs revealed very different opinions on how the SPP should be organized to best suit their work routines:

"It should look like the paper-based requisition we have today. Just get it up on the screen and click all over the form". (General Practitioner 1)

"There are too many [analyses]. It should be simplified and packages of analyses to choose from should be made". (General Practitioner 2)

The SPP editors intended to take the different views into account and organize the list in an intuitive way for the GP. The first try was a mix between analyses sorted by clinical criteria, frequency of use and sorting according to laboratory executing the analyses. After presenting this version to the GPs the reactions from the GPs were still diverging, and it was quite clear that what was most obvious for one GP was not obvious or intuitive for another. A screen shot of the Well Interactor window, showing the SPP in the left part of the window is shown below, Figure 2.

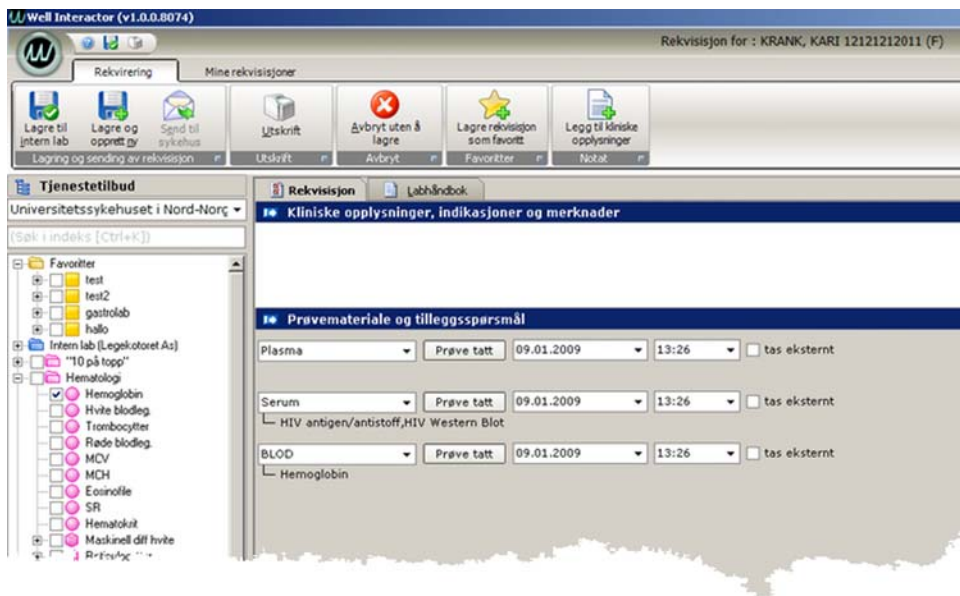


Figure 2. Screen shot of Well Interactor smart client.

To make it more generic and to encompass the diverging work practices among the GPs Well Diagnostics developed a feature to enable the GPs to define their own “Favourites”. This made it possible for each GP to adapt the system more to his or her individual work practice by gathering analyses that were often ordered together as a package. But it also became clear that the other user groups in general practice, the receptionists and laboratory personnel, had other needs regarding the list of analyses. They only order particular routine analyses and do not make the decisions on which analyses to choose, meaning that sorting the list according to clinical criteria would not make sense for them. An alphabetical searchable list would be more useful.

Well Diagnostics considered the GPs’ satisfaction with the SPP as crucial for their overall satisfaction with Well Interactor. Even though the structuring of the SPP was up to each hospital to decide, and was not part of the product as such, the vendor realized that it needed to be involved in this process to ensure effective cooperation between the hospitals and the GPs in establishing the SPP. There had previously only been occasional contacts between the laboratory and the GPs, and the vendor thus had to help establish a relationship:

“The laboratory wants one thing, and the GP wants something else. Then we arrange an informal meeting with pizza and beer; or we get the laboratory to visit the GP practice. They discuss and compromise.” (Designer, Well Diagnostics)

One example was the structuring of the microbiology analyses in the SPP at UNN. The microbiology laboratory and the pilot GP practice disagreed on how this should be done. Well Diagnostics arranged a meeting first with the GPs where the GPs presented a simplified structure with rarely used analysis made less visible. A sketch was drawn and the developer brought the sketch to the microbiology laboratory. The proposal was met with scepticism because the biomedical laboratory scientists were afraid that the rarely used analysis would be too difficult to find, thus preventing the requisitioner to order the most appropriate analyses. The vendor then arranged a joint meeting where the laboratory and the GP practice managed to agree on a SPP to be tried out at the GP pilot site.

Summing up; to meet these contradictory requirements for the SPP two strategies were chosen to ensure flexibility on both sides, to make use of “Favourites” for the GPs and to create the SPP Publisher for the laboratory personnel. In addition the vendor chose to become involved in the laboratories’ work to organize the SPP by facilitating communication between the user groups. Because of the practical problems of involving the actual users in primary care, and also because UNN as the first customer lacked contact with the users in primary care, the developers at Well Diagnostics had to act as mediators and user proxies.

5.2.2. Different laboratory systems—integration challenges

The microbiology laboratory at UNN uses a different laboratory information system than the clinical biochemistry laboratory. This also resulted in new

challenges related to integration. As mentioned earlier the microbiology laboratory was able to export most information whereas the pathology laboratory system at UNN which will soon be integrated is only capable of exporting some of the information needed. To make up for this, Well Diagnostics will develop a small software component to accommodate for the missing export functionality by transforming the existing data into the needed formats. The communication in the Well Interactor concept is based on EDI messaging. Not all the laboratory systems have the possibilities to import such standardized messages. The planned integration with the pathology laboratory information system is put on hold waiting for the implementation of this functionality in the pathology system. The microbiology information system had this functionality, but the national standard for microbiology requisitions were even more deficit than for the clinical biochemistry requisitions, making it necessary to define another API together with the vendor of the microbiology information system.

A larger challenge occurred when the requisitions were imported into the microbiology information system. This system was not able to handle identification of samples in the same way as the clinical biochemistry laboratory's information system. This required workarounds such as double or manual registration in the receiving of requisitions and samples. Another problem was the complexity of the microbiology information system which handled internal logistics, analysing results, storage information and sending of results back to the GPs. All these parts of the system were somehow influenced by the incoming electronic requisitions, and unanticipated changes and errors occurred, and had to be amended by the vendor of the microbiology information system before electronic requisitions for microbiology laboratories could be implemented for routine use.

The dependency on vendors of GPs' EPR systems and laboratory information systems has presented important challenges related to integration:

"...they have their own agendas and their own products. And we, who have to make it work together, are often dependent on their priorities. That is the problem; to get a reaction from the vendors. (..)The more vendors you need help from the longer it takes to complete. It is a consideration of costs (...) it is difficult to get instant commitment from other vendors." (Designer, Well Diagnostics)

This has caused delays in the development of Well Interactor, because the vendors of the laboratory information systems have been delayed with the components that were needed for using electronic requisitions. Well Diagnostics has invested much time and effort in the cooperation with these vendors to ensure progress in the development process.

The challenges related to integration are partly due to the fact that Well Interactor is the first commercially available solution for electronic requisitions in Norway. The work related to making standards is often slower than the process of making working solutions. For that reason the contract between the GPs' EPR and the smart client will probably serve as a de facto standard for other vendors

that want to include the possibility of electronic requisitions in their EPR system. To be first in the market creates opportunities and may constitute a monopoly, but it also forces the vendor to make decisions that will have long-term consequences related to product development.

5.3. Moving out of the local context (2006)

5.3.1. *A new setting for the product*

In June 2006, while Well Diagnostics was in the process of designing and developing the first version of Well Interactor, they won a contract for electronic requisitions at the Akershus University Hospital (Ahus). Ahus is among the largest hospitals in the southern part of Norway, serving a densely populated area around Oslo. For Well Diagnostics, this represented a breakthrough in this particular market and an important step towards commercialisation of Well Interactor. Being part of an agile development process, Ahus bought what had been developed so far at the time of purchase, and has taken part in the ongoing design process of the system. After the implementation of Well Interactor at Ahus and surrounding general practices the vendor has experienced a new setting for the product. For instance Ahus had only one laboratory information system for the whole hospital and a fully automated pre-analytic unit where the samples and the electronic requisitions could go straight into the system. The developer described how he observed the differences from UNN in this way:

"For instance at Ahus there has been great enthusiasm throughout the entire organisation to make this work well.[...] Their pre-analytic unit was to be fully automated. At UNN there is another story. There the receiving of samples is done manually; they re-label samples on arrival and new distributions. Much more labour intensive." (Designer, Well Diagnostics)

5.3.2. *New customers; new requirements and relations*

Even though going to Ahus represented an easier case than UNN in many respects, the prioritisation of user stories took on a new dimension as Well Interactor became a commercial system. For instance, faced with choices between new user stories entailing new functionalities and mere improvements of existing ones, those promising the most cash flow were likely to win:

"....when we set priorities (...). If a receptionist requests one user story, and another user story requires the same working effort, but gives 200 000 [Norwegian kroner] incoming cash flow, then the user story with the cash flow unfortunately gets a higher priority. We have to balance that all the time (...). It is business also". (Designer, Well Diagnostics)

The sales to Ahus and other hospitals have resulted in large numbers of new user stories to be prioritized by the development team. Some of these might have general interests for many customers, while others might be based on the specific needs of one particular user. For instance Ahus wanted the possibility to back order analyses or modify requisitions that had already been sent. This was not part of the existing Well Interactor. UNN on the other hand was not interested in this functionality. The strategy of Well Diagnostics became to divide the product into modules; a basic module, a status and back ordering module and also a module for forums and dialogue.

In the beginning the vendor did not need a systematic method for collection and refinement of user stories, or a well-defined strategy for sifting and smoothing of the general and the particular stories before they went to the design team for prioritising. As the amount of user stories grew, the need for systematic methods emerged. One of the strategies was to arrange a workshop for the SPP editors from the different hospitals to get feedback from the users and to facilitate exchange of experience among the users. As a result of this, users from different hospitals established relationships, and started to exchange experience outside the workshop as well.

In the project phase the users were able to contact the designers or marketing people directly with feedback or needs. When more customers appeared one strategy was to shift more of the communication with the customers from the development team towards the support and marketing personnel. Some hospitals, among them Ahus, have taken on the user proxy role for the primary care users themselves by mediating and also taking a stand on issues raised by their general practice users.

5.4. Future prospects: Beyond laboratory requisitions

The generification to new customers has taken the product towards a more module-based product, making it possible for customers to purchase those parts of the product that satisfy their needs. In the future this will be taken even further to include other services that require interaction between general practice and the hospitals. Presently Well Diagnostics is in the first phase of extending the product to include interactive referrals. These orders include processes similar to those for laboratory requisitions at the GP practices, with the necessity to transfer information automatically from the EPR. But these orders will require different content from the GPs and the function will involve different user groups as well as different information systems in the hospitals.

6. Discussion

In the discussion, we will a) focus on the negotiations between Well Diagnostics and the vendors; b) we elaborate on the strategy of aligning the user communities

as well as creating a more generalized functionality, and c) we indicate how the roles of the involved actors have changed during this process.

6.1. Negotiating the boundaries of an integrated system

Several studies underscore how boundaries are intrinsic to organizations (see Hernes 2004; Kerosuo 2004; Barrett et al. 2007). A key point in this literature is how boundaries are continuously produced and reproduced, countering claims in the management literature related to boundaryless organizations (Ashkenas et al. 1995) or clear-cut boundaries to support a seamless information and process flow (Davenport 1993). The argument of sliding boundaries is also used by Pollock and Williams (Pollock and Williams 2008) when they elaborate on how generic systems are results of complex boundary work between vendors and users.

Moving beyond this, we would like to draw attention to the generification process of systems intended for integration with larger portfolios, and particularly how this presupposes drawing and redrawing of the boundaries between the vendors involved. Specifically, we analyse the interactions between Well Diagnostics and the vendors of the larger systems, most notably Profdoc and DIPS ASA. The basic core of Well Interactor's functionality was associated with its integration and dependency of the GPs' EPR in primary care and the laboratory information systems in the hospitals. This forced Well Diagnostics to engage in extensive negotiations and clarifications with these vendors.

It was particularly important to integrate with Profdoc, which by controlling the major part of the primary care market acted as a gatekeeper (Latour 1987) to access the GPs. On the other hand, Profdoc was only moderately interested in the new product because, as argued by a key informant from Profdoc, *"this would add functionality that only benefited the users in the laboratories and not the GPs"*.

Well Diagnostics was aware that it was negotiating with Profdoc from an inferior position. However, Well Diagnostics' advantage was that cross-organizational collaboration in healthcare was high on the political agenda in Norway (Ministry of Health and Care Services 2006), making it evident that some sort of Government-initiated integration project would be launched in any case. For Profdoc then, working together with Well Diagnostics would enhance control of such development, as the vendors had a history of collaboration.

Another factor that kept Profdoc aligned was the promise made by Well Diagnostics to develop a product that did not involve a substantial proportion of Profdoc's development resources. Consequently, if the new product failed, the consequences for Profdoc would be limited. In addition, Well Diagnostics sweetened the pill by offering Profdoc shares of the income from sales to the hospitals.

However, Well Diagnostics knew that an unforeseen shortage of resources due to for instance new sales among the other vendors (not only Profdoc) could suddenly alter their priorities. This was a recurring problem, as pointed out by one of the marketing persons in Well Diagnostics: *"The functionalities in our systems depend to*

a large degree on what the vendors of the electronic patient records systems will do.”

This created a situation where it was mainly up to the larger vendors to define the conditions for the integration, that is, how and when the integration should occur and what should be integrated. It was a so-called asymmetrical integration (Sahay et al. 2009). Well Diagnostics willingly went far beyond what could be expected by doing work that would naturally have been associated with the other vendors. For instance, one of the laboratory systems at UNN lacked the ability to export information to the SPP editing tool. Instead of waiting for the vendor of this system to make it possible within the laboratory system, Well Diagnostics undertook and accomplished the task.

Interestingly, during the course of the project the situation changed, and Well Diagnostics managed to establish more manoeuvrability for itself by expanding (Gieryn 1999; Barrett et al. 2007; Pollock and Williams 2008) the boundaries around Well Interactor. This occurred in three different ways. Firstly, as the new system was well received and sold to several hospitals, the boundary work between the vendors increasingly became shaped by demands from users. And they demanded functionality that increasingly included work done by Profdoc. One example was how the users forced Profdoc to implement a non-standard integration protocol because the users wanted to order microbiology and clinical biochemistry on the same requisition, while the national standard specified separate requisitions. Another example was that Profdoc made it possible for the Well Interactor referral window to appear as part of the referral template in the EPR because they realized that users would appreciate this functionality.

Secondly, Well Diagnostics had initially planned to avoid significant involvement in complex organizational issues inside the hospital, but later realized that such involvement would be necessary for effective integration of the different laboratory systems. The positive effect of this was that when the developers understood how things worked, they could see things from the hospital users' perspective (Boland and Tenkasi 1995). This 'inside' knowledge was fed into the negotiation process with the other vendors. For instance, Well Diagnostics tried to influence DIPS ASA, the vendor of the laboratory system used in the pre-analytic unit, to change their priorities in order to develop more efficient routines for handling electronic requisitions in the hospital.

Thirdly, it appeared that Well Diagnostics was increasingly taking on the role of a go-between or a boundary organization (Guston 2001) between the larger vendors. Initially the situation was more unsettled, since Well Diagnostics as well as both Profdoc and DIPS ASA were involved in the domain with different products, and the boundaries between the vendors were not always clearly defined. However, by mediating between the vendors, Well Diagnostics was actually able to clarify these boundaries.

6.2. Aligning the users of an agile product

Often vendors want to reuse much of their software code in order to optimize payback and to manage different implementations at different sites in a relatively

similar way. It is therefore imperative for vendors to carefully balance the boundary between the particular and the general functionality while at the same time trying to expand (Gieryn 1999, p.15–17; Barrett et al. 2007) the general part of it. A basic strategy is to try to align the user communities in order to obtain common requirements (Pollock and Williams 2008).

After implementation at several hospitals, the number of users of Well Interactor increased. Well Diagnostics therefore aimed at moving user requirements to a public arena. The vendor organized a workshop for the SPP editors from the different hospitals. The intention of the workshop was twofold: to get feedback from the users and to facilitate exchange of experience among the users. Interestingly, as the participants got to know each other they started to consult each other on decisions they had made, for example: *“What do you think of a character limit of n characters for this field?”* Another example was a discussion on the size and the appearance of the labels to be glued to the tubes. The different hospitals used them in different ways and the participants ended up agreeing to send examples to each other, in order to exchange ideas. The observed coordination among the users themselves is in line with the findings of Pollock et al. (2007).

However, Well Diagnostics found that there were limitations to how far it was possible to align the users. The work processes were so divergent among the different laboratories as well as between the hospitals that a generic version was not desirable. According to agile philosophy, satisfying the customer should always have the highest priority, thus making the vendor reluctant to implement uncompromising alignment strategies. Instead the vendor created the SPP Publisher, a tool making it possible for each hospital to make and maintain its own list of services. This enabled the users in the hospitals to experiment with how they presented their services to the GPs. In this way, the SPP Publisher also represented a generification, as this tool enabled easy tailoring to local needs and work practices.

Provided with the SPP Publisher, the users of the different sites gradually learned from each other or started to consult and use each other's SPPs. For instance, the laboratory at UNN found the SPP designed by Ahus to be quite useful:

“We got the service provider profile from Ahus. [...] Our organization of the SPP was not good, [...]. Now we have made great changes in the SPP and borrowed from the way Ahus has organized their SPP.” (Biomedical laboratory scientist, UNN).

The same SPP from Ahus was also used by a third hospital in the south of Norway when its users were learning to create the SPP. Later it was used as the point of departure when they started to make their own SPP. But as this customer expressed it quite humorously, they would not use it as is, but would rather use some of its ideas:

“We would like to look at it [the SPP from Ahus], but we want to make our own. We want to ‘re-invent the wheel’ ourselves”, (Biomedical laboratory scientist, Asker and Bærum Hospital)

This biomedical laboratory scientist thus emphasized that their work practice was different from work practices in the other hospitals. Nonetheless, in reality they learned a great deal from the SPP from Ahus, and in this way involved themselves in the generification process.

The vendor Well Diagnostics has invested much time and effort on issues related to the structuring of the SPP for the pilot sites. The obvious reason for this has been that user satisfaction with the system is closely related to how the SPP is structured and specified, and the vendor thus wanted to enforce user satisfaction also on that part of the system. By engaging in meetings with the separate user groups, and facilitating meetings between the users in primary care and in the laboratories, the vendors have forced the users to actually cooperate on the design of the SPP. Interestingly, it seems that the spirit of agile methodologies has spread to the users in the laboratories as well, in their iterative approach to the SPP design. The first users at UNN started by trying to make the transition from paper-based to electronic requisitions as smooth as possible:

“We took the colouring of our paper requisitions as a starting point, and tried to make it as identical as possible. We have got feedback that it was OK, but that they [the GPs] had not really seen that connection [to the paper-based requisitions]”. (Biomedical laboratory scientist, UNN)

In accordance with agile methodology they took the feedback they received from this first version, and made small changes to the SPP, which was continually downloaded to the users in primary care. This iterative process has continued as they are trying to find a good solution.

6.3. Redressing the users as customers

In agile software development methods such as Extreme Programming and Scrum the role of the customer is well defined while the role of the actual users is less prominent, and the methods do not specify how to involve the users in the process (Rittenbruch et al. 2002). In the development of a standalone system, and systems made for one particular customer, the customer role might be straightforward. In a large integrated setting, on the other hand, the situation becomes more complicated; as the user roles may become mixed up in a set of other roles, especially since the product goes through continuous evolution. For the vendor it seemed at first that the user groups in general practice and in the laboratories were well defined and clear cut because they belonged to different organizations with different responsibilities. But it became evident that when working with systems that involved both parties the boundaries were not that clear and especially not when it came to making decisions. One example was the work of structuring the microbiology part of UNN's SPP, which involved much mediating and 'run between' for the vendor.

In the early phases of Well Interactor the developers worked closely with the GPs. User stories were established, and GPs were even paid by Well Diagnostics to participate in the project. However, as the system was sold to more hospitals, the user role gradually changed into a customer role. This had two implications: The expectations in relation to the users/customers were not agreed upon, and for the vendor, it raised the question of how to represent the user in the agile design process.

For the initial users at UNN, this transition to a different role was challenging. In the beginning they were part of the project, representing the users in the hospital. Later on as the project phase was completed, these users were transformed into regular customers. As the market for Well Interactor started to grow, the users at UNN were not the only customer for the product, but one of several that had to fight to be heard, and they felt that they did not get the same responsiveness from Well Diagnostics. One example was wishes for functionalities regarding tailoring to their complicated routines for the receiving of samples:

“(..) when our requirements were in conflict with how Well Diagnostics thought it ought to be, I felt that they chose to do it their way. Because they wanted it to be a product to be used other places.” (Biomedical laboratory scientist, UNN)

Moreover, it was not clear for the users in the hospital what the new customer role implied for them. As customers they owned the system and they had to distribute it to, and mobilize the users in primary care. However, this was not initially obvious to these users:

“For us this has been unclear. [...] We thought of it as Well Diagnostics’ product, and I think we would have demanded to be more involved if we had known that we had to take it further [to the users in primary care]”. (Biomedical laboratory scientist, UNN)

This ambiguity has resulted in delays of the mobilization of users in primary care since the users at UNN had to attend to other internal matters first.

Another important implication of the transition from users to customer contacts was that the customers did not have a basis in the group of end users, the GPs. Instead, the GPs were represented by intermediaries (Pollock and Williams 2008) such as customer representatives from the hospital. By passing on requests, redefining requirements, and taking a stand on what requirements and wishes should be passed on to the developers, the customers controlled what the actual users (the GPs) were allowed to have. The following quote from a customer representative is illuminating:

“We had a GP practice that wanted the possibility of printing one label for each analysis. For example, one for haemoglobin, one for CRP, etc. That would make it much easier for them because they then could label all tubes with these labels and also write the results on a duplicate label before entering it into the electronic patient record system. But I took the liberty of saying no

to them. [...] I asked them: how did you do it before? [...] The labels are expensive, and as long as we provide the GP practices with labels and printers they will not get one label for each tube for internal use.” (Contact person between the hospital and GP practices, Ahus)

Although the vendor did not distrust this customer representative the vendor still felt the urge to have some informal contact with some of the users in general practice through site visits and telephone to get direct feedback from them.

The use of the marketing personnel as intermediaries in the development team increased in line with the increasing number of customers. The marketing personnel stayed in continuous contact with customers by telephone, by site visits, and through information and training sessions. The acquired feedback was fed into the planning and prioritization process in the development team to enable the continuous improvements and new releases that are characteristic of agile methods. In this way the marketing personnel replaced the developers' close contact with the actual users.

It is evident that the use of intermediaries increases as the customer role is more prominent, and that many different actors can serve as these representatives. For the vendor this implies a user base that is easier to handle, but might introduce the risk of filtering and distortion of information (Keil and Carmel 1995), and might need to maintain the close contact with users in other ways.

7. Conclusion

This study shows that the bottom-up approach to developing systems clearly may emerge successfully. Well Interactor is now in a process of evolving from a small tailored and integrated product for one customer to a product for a larger market. During this generification process the vendor has employed diverse strategies towards other vendors as well as towards the users. The aim of this study has been to contribute to strategies to vendors who want to move their locally designed and highly integrated systems to a larger market. A further aim has been to explore how such systems developed for a local practice, and tightly integrated with the existing infrastructure, can be adapted to a larger market. We have drawn on the notions of generification and boundary work to describe this process. We have found that generification in a highly integrated environment raises the important issues of boundary work towards other vendors and towards the different heterogeneous user groups.

We have shown that in an integrated context the boundaries between the different actors have not been entirely obvious, due to the emerging character of the system. For the vendor this has implied continuous negotiations towards the other vendors. We also found that when the user groups belonged to different organizations, and particularly when some of these groups were not the customer of the product, this implied more demanding boundary work between the vendor

and the users than otherwise would have been needed. We have also shown that the transition from working with the actual users to working with customer representatives might be challenging and confusing for all actors. Accordingly, to obtain success, it may be necessary for vendors to establish strategies to maintain contact with the end-users.

The strategic implication of our findings for a small vendor is the need to engage in boundary work from the start, to anchor the product and find its place in the landscape with the larger vendors. This requires much patience as the vendor depends so much on the priorities of the other vendors. It also implies that the vendor must expect to engage in the boundary work with the different users groups. This work is tedious and will require that the vendor invest much time and effort by trying to mobilize, and sometimes 'running between', the other actors in the process.

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